

PRD — Referral Optimizer for Primary Care

1. Product Summary

Working name: Referral Compass

One-liner: An AI referral-matching website that helps family physicians choose the right specialist, avoid failed referrals/eConsults, and auto-fill referral forms from minimal input.

Core user input:

1. Referral reason, preferably dictated by microphone
2. Patient location
3. Accessibility/language comments
4. Optional: scribe transcript, EMR extract, physician saved preferences

Core output:

1. Ranked list of optimal specialists/services
 2. Explanation of fit, wait time, travel burden, accessibility, language match, and estimated triage priority
 3. Completed synthetic Ocean-style referral form ready for review/export
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2. Problem

Family physicians lose time because referral workflows require them to:

- Know each specialist's true clinical scope
- Search directories manually
- Guess wait times and triage categories
- Re-enter EMR data into forms
- Recover from rejected or misdirected referrals
- Remember where they usually refer
- Account for patient geography, mobility, language, and accessibility needs

Failed referrals and failed eConsults waste clinician time, delay care, and create avoidable administrative loops.

3. Target Users

Primary

Family physicians and clinic staff in Ottawa/Ontario.

Secondary

Specialist offices, centralized intake teams, eConsult coordinators, and referral management teams.

4. Hackathon MVP Goal

Build a working demo that takes a short referral reason and patient context, ranks specialists from synthetic OceanAPI data, and generates a completed referral form with transparent rationale.

MVP promise:

"From 30–60 seconds of dictation to a ranked referral destination and a filled form."

5. Awards Alignment

Privacy, safety, and trust

- Synthetic data only for hackathon
- No persistent storage by default
- Optional saved physician preferences
- Explicit consent for uploaded transcript/EMR data
- Human-in-the-loop final review before form submission
- Explainable ranking, not black-box routing

Primary care workflow fit

- Minimal required input
- Voice-first referral reason
- No forced login for basic use
- Form filling happens after recommendation
- Saved "usual referral patterns" only if physician opts in
- Output is reviewable and exportable

Creative / technically impressive AI

- AI scope matching between referral reason and specialist/service scope
 - Triage likelihood estimation
 - Missing-information detection
 - Form auto-completion from transcript/EMR text
 - Learning loop from opt-in referral success/failure feedback
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6. MVP Features

6.1 Referral Intake

Required fields:

- Referral reason: free text or voice
- Patient postal code / city
- Maximum travel distance
- Accessibility comments
- Language preference, especially French
- Optional urgency flag

Optional fields:

- Scribe transcript upload
- EMR data extract
- Usual referral provider
- Physician saved preferences

6.2 Specialist Ranking

The system returns a ranked list:

Rank Factor	Description
Scope match	Does the provider actually see this condition?
Wait time	Estimated days/weeks to consult
Geography	Distance from patient location
Triage fit	Likely triage category based on symptoms
Accessibility	Wheelchair access, parking, transit, virtual option
Language	French-speaking availability
Referral success rate	Synthetic historical acceptance rate
Physician preference	Optional saved “usual referral” boost

Output example:

1. **Ottawa Digestive Health — Dr. Leclerc**

2. Scope match: 94%

3. Wait estimate: 6 weeks
4. Travel: 4.2 km
5. French-speaking: yes
6. Triage estimate: semi-urgent

7. Why: accepts iron-deficiency anemia + alarm GI symptoms; high acceptance rate for similar referrals

8. Eastern Ontario GI Centre — Central Intake

9. Scope match: 89%

10. Wait estimate: 4 weeks
 11. Travel: 11.8 km
 12. French-speaking: partial
 13. Why: faster access but weaker language/accessibility fit
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6.3 Form Auto-Fill

The app generates a synthetic Ocean-style referral form using:

- Referral reason
- Extracted clinical details
- Patient demographics from synthetic EMR data
- Medication list
- Relevant labs/imaging
- Red flags
- Accessibility/language notes
- Requested service
- Triage-relevant fields

The physician can review and edit before export.

6.4 Missing Information Detection

The model flags missing data that could cause rejection or triage downgrading.

Example:

“This GI referral may require hemoglobin, ferritin, FIT/FOBT result, medication list, and prior colonoscopy history. Ferritin is missing from the uploaded EMR extract.”

6.5 Learning Loop

After referral outcome, opt-in physicians can mark:

- Accepted
- Rejected
- Redirected
- Wrong scope
- Too long wait
- Patient declined due to travel/accessibility
- Triage priority changed

This improves future scope matching and ranking.

No identifiable patient data is needed for learning; only structured outcome features are stored.

7. Data Structures

7.1 Patient

```
{
  "patient_id": "synthetic_patient_001",
  "age": 54,
  "sex": "F",
  "postal_code": "K1N",
  "city": "Ottawa",
  "language_preference": "French",
  "accessibility_needs": ["wheelchair_access", "near_transit"],
  "max_travel_km": 15
}
```

7.2 Referral Request

```
{
  "referral_id": "ref_001",
  "physician_id": "fp_123",
  "referral_reason": "Iron-deficiency anemia, unintentional weight loss,
dyspepsia. Needs GI assessment.",
  "urgency_hint": "semi_urgent",
  "comments": "Patient prefers French-speaking provider and wheelchair-
accessible site.",
  "uploaded_sources": {
    "scribe_transcript": true,
    "emr_extract": true
  }
}
```

```
}  
}
```

7.3 Specialist / Service

```
{  
  "provider_id": "spec_045",  
  "name": "Dr. Amelie Leclerc",  
  "specialty": "Gastroenterology",  
  "clinic": "Ottawa Digestive Health",  
  "location": {  
    "city": "Ottawa",  
    "postal_code": "K1K",  
    "lat": 45.44,  
    "lng": -75.65  
  },  
  "languages": ["English", "French"],  
  "accessibility": ["wheelchair_access", "near_transit"],  
  "accepted_conditions": [  
    "iron_deficiency_anemia",  
    "dyspepsia",  
    "weight_loss",  
    "suspected_gi_malignancy",  
    "ibd"  
  ],  
  "excluded_conditions": [  
    "functional_abdominal_pain_without_red_flags"  
  ],  
  "average_wait_days": 42,  
  "urgent_wait_days": 14,  
  "acceptance_rate": 0.91,  
  "usual_referral_score": 0.72  
}
```

7.4 Referral Form Template

```
{  
  "form_id": "ocean_gi_synthetic_v1",  
  "specialty": "Gastroenterology",  
  "required_fields": [  
    "reason_for_referral",  
    "red_flags",  
    "relevant_labs",  
    "medications",  
    "prior_imaging",  
  ],  
}
```

```

    "prior_endoscopy",
    "urgency",
    "patient_accessibility_needs"
  ],
  "triage_fields": [
    "weight_loss",
    "anemia",
    "gi_bleeding",
    "dysphagia",
    "family_history_crc"
  ]
}

```

7.5 Ranking Output

```

{
  "provider_id": "spec_045",
  "rank": 1,
  "overall_score": 0.92,
  "scope_match_score": 0.94,
  "wait_score": 0.81,
  "distance_score": 0.88,
  "accessibility_score": 1.0,
  "language_score": 1.0,
  "triage_fit_score": 0.9,
  "preference_score": 0.72,
  "explanation": [
    "Accepts iron-deficiency anemia and alarm GI symptoms",
    "French-speaking provider",
    "Wheelchair-accessible site",
    "Estimated semi-urgent triage"
  ]
}

```

7.6 Referral Outcome Feedback

```

{
  "referral_id": "ref_001",
  "provider_id": "spec_045",
  "outcome": "accepted",
  "triage_category": "semi_urgent",
  "actual_wait_days": 38,
  "reason_if_failed": null,
}

```

```
"opt_in_learning": true
}
```

8. AI Flow

Step 1 — Parse referral reason

Input:

"54-year-old woman with iron-deficiency anemia, dyspepsia, and 10-pound unintentional weight loss. Needs GI."

Model extracts:

```
{
  "specialty_intent": "gastroenterology",
  "conditions": ["iron_deficiency_anemia", "dyspepsia", "weight_loss"],
  "red_flags": ["weight_loss", "anemia"],
  "urgency": "semi_urgent",
  "required_form_data": ["CBC", "ferritin", "medications",
    "prior_scope_history"]
}
```

Step 2 — Match scope

Compare extracted clinical intent against each specialist's accepted/excluded conditions.

Step 3 — Estimate triage

Use synthetic rules:

- Red flags → semi-urgent or urgent
- Stable chronic issue → routine
- Incomplete referral → lower confidence

Step 4 — Rank providers

Suggested scoring:

Ranking is not based on a simple weighted formula.

Instead, providers are filtered and ranked through a staged decision process:

1. Scope Match (Required Threshold)
 - The provider must accept the referral reason and clinical condition.
 - Providers that explicitly exclude the condition are removed.
 - Partial scope matches may still be considered if no ideal match exists, but are ranked lower and clearly labeled.
2. Accessibility and Patient Constraints (Required Thresholds)
 - Patient requirements such as wheelchair accessibility, transit access, virtual care availability, language preference (e.g., French-speaking), and maximum travel distance are evaluated.
 - Providers that do not meet mandatory patient requirements are deprioritized or excluded depending on the constraint.
 - Strong accessibility and language matches are surfaced prominently.
3. Triage and Clinical Fit
 - The referral reason is analyzed for urgency, red flags, and likely triage category.
 - Providers whose intake criteria and service model align with the expected triage level are prioritized.
 - The system also identifies referrals that may be missing information required for appropriate triage.
4. Wait Time Optimization (Primary Ranking Factor)
 - Once scope, accessibility, and triage requirements have been satisfied, estimated wait time becomes the primary differentiator between providers.
 - Among clinically appropriate options, providers with shorter expected waits are ranked higher.
 - Urgent and semi-urgent referrals place even greater emphasis on timely access.
5. Secondary Tie-Breakers
 - Geographic proximity
 - Historical referral acceptance rates
 - Physician referral preferences (if enabled)
 - Patient convenience factors

Each recommendation includes a transparent explanation describing why it was ranked, which requirements were satisfied, and any trade-offs between wait time, distance, accessibility, or provider fit.

Step 5 — Fill form

Use extracted EMR/transcript values to populate fields.

Step 6 — Explain and export

Show ranked providers and generated form with editable sections.

9. Privacy and Safety Design

- Synthetic data only for hackathon
 - No PHI required for demo
 - Local session mode: no data stored
 - Optional login only for preferences
 - Opt-in feedback only
 - Clinician must approve final referral
 - AI recommendations are explainable
 - Missing data and confidence displayed
 - Audit-style referral rationale generated
 - No autonomous submission in MVP
-

10. Demo Use Cases

Demo 1 — “The almost-failed GI referral”

Scenario:

A family doctor dictates:

“Patient in Vanier, prefers French, wheelchair access. Iron-deficiency anemia, dyspepsia, 10-pound weight loss. Needs GI.”

Demo arc:

1. Voice input transcribes referral reason.
2. App detects alarm features.
3. App ranks GI specialists.
4. App avoids a provider who excludes suspected malignancy workups.
5. App recommends a French-speaking, wheelchair-accessible GI clinic with strong acceptance history.
6. App auto-fills the synthetic GI form.
7. App flags missing ferritin.
8. Physician clicks “Add ferritin from EMR.”
9. Completed form is ready.

Pitch moment:

“Instead of asking the physician to know every specialist’s hidden scope rules, the system reads the clinical intent and matches it to the right door.”

Demo 2 — “The closest specialist is not always the best referral”

Scenario:

Referral reason:

“Chronic abdominal pain and iron-deficiency anemia. Patient lives in Orleans, prefers French, uses a wheelchair, and can travel up to 20 km.”

Demo arc:

1. The physician dictates the referral reason and enters the patient's location and accessibility preferences.
2. The system identifies gastroenterology as the requested specialty based on the referral reason, without making any diagnosis or clinical determination.
3. The nearest GI clinic appears first in a simple distance sort, but the system explains why it is not the optimal referral:
4. Limited wheelchair accessibility
5. Does not routinely accept referrals matching this referral reason
6. Longer actual wait times reported through opt-in feedback than publicly listed
7. A slightly farther French-speaking clinic is ranked higher because it meets accessibility requirements, has a stronger historical acceptance rate for similar referrals, and has a shorter observed wait time.
8. The physician can view transparent reasoning for each recommendation, including scope fit, accessibility fit, language fit, travel burden, and wait-time considerations.
9. The referral form is automatically completed using the referral reason and optional uploaded transcript/EMR data.
10. The system highlights any missing fields that may delay processing and allows the physician to review before export.

Learning loop demonstration:

1. After the referral outcome, a physician optionally reports:
2. Referral accepted
3. Actual wait time was 5 weeks instead of the listed 10 weeks
4. Multiple physicians provide similar feedback over time.
5. The ranking engine gradually improves its wait-time estimates and referral recommendations using aggregated, non-identifiable outcome data.
6. During the demo, a "before vs after learning" view shows how the same referral would have been ranked using directory data alone versus using community-reported outcomes.
7. The audience sees that recommendations improve not because the system diagnoses patients, but because it learns which referrals are accepted, how long patients actually wait, and which providers are the best fit for specific referral requests.

Pitch moment:

“Most referral tools stop at finding the nearest specialist. We help physicians find the specialist most likely to accept the referral, meet the patient's needs, and see them sooner—while continuously improving through opt-in real-world referral outcomes.”

Demo 3 — “Patient-centred routing”

Scenario:

Referral reason:

“Older patient with cataracts. Lives in Orleans, cannot drive, prefers French.”

Demo arc:

1. App ranks ophthalmology options by travel burden, French access, and wait time.
2. A slightly longer-wait clinic ranks higher because it is transit-accessible and French-speaking.
3. Physician sees the tradeoff transparently.

Pitch moment:

“The fastest referral is not always the best referral. The best referral is the one the patient can actually complete.”

11. Demo Screen Flow

Screen 1 — New Referral

- Referral reason text box
- Microphone button
- Patient postal code
- Max travel distance
- Accessibility/language notes
- Optional upload: transcript / EMR extract

Screen 2 — AI Clinical Parse

- Suspected specialty
- Key conditions
- Red flags
- Triage estimate
- Missing data

Screen 3 — Ranked Specialists

Cards showing:

- Specialist/service name
- Scope match
- Wait time
- Distance
- Language/accessibility match
- Acceptance likelihood

- “Why this ranked here”

Screen 4 — Filled Referral Form

- Editable synthetic Ocean-style form
- Highlighted auto-filled fields
- Highlighted missing fields
- Export button

Screen 5 — Outcome Feedback

- Accepted / rejected / redirected
 - Reason if failed
 - Opt-in learning toggle
-

12. Success Metrics

Hackathon demo metrics

- Time from referral reason to completed form
- Number of clicks saved
- Number of missing fields detected
- Correct specialty selection in demo cases
- Explainability of ranking

Product metrics

- Referral rejection rate
 - Referral redirection rate
 - Time to completed referral
 - Physician satisfaction
 - Patient travel burden
 - Specialist acceptance rate
 - Triage accuracy
-

13. MVP Non-Goals

- Real clinical deployment
 - Real PHI storage
 - Autonomous referral submission
 - Replacement of physician judgment
 - Real-time integration with production Ocean or EMR systems
 - Definitive medical triage decisions
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14. Build Plan

Frontend

- React / Next.js website
- Voice input
- Referral wizard
- Ranked specialist cards
- Editable referral form

Backend

- Synthetic OceanAPI service
- Provider directory endpoint
- Form template endpoint
- Referral ranking endpoint
- Feedback endpoint

AI Components

- Referral reason parser
- Scope matcher
- Triage estimator
- Missing-field detector
- Form filler
- Explanation generator

Synthetic API Endpoints

```
GET /providers
GET /providers?specialty=gi&city=ottawa
GET /forms/:specialty
POST /referrals/parse
POST /referrals/rank
POST /forms/fill
POST /feedback
```

15. Final Pitch

Referral Compass reduces failed referrals by matching the physician's clinical intent to the right specialist, right form, right triage pathway, and right patient constraints.

It does not ask family doctors to do more data entry. It lets them say the referral reason out loud, checks scope and triage fit, fills the form, and keeps the clinician in control.

The core insight:

Primary care does not need another directory. It needs a referral co-pilot that understands clinical scope, local access, patient constraints, and the hidden reasons referrals fail.